

TOURNIQUET (TQ) DOWNGRADE and CONVERSION TO WOUND PACKING DRILLS

DOWNGRADE CRITERIA

HAEMODYNAMICALLY STABLE, WOUND PACKING RESOURCES and MONITORING AVAILABLE
TQ IS NOT NECESSARY or CAN BE APPLIED MORE EFFICIENTLY
SHOULD BE CONSIDERED IN PROLONGED CASUALTY CARE SETTINGS >2 HOURS

DOWNGRADE ACTIONS

1. ENSURE APPROPRIATE AND SUFFICIENT ANALGESIA/SEDATION PRIOR TO DOWNGRADE.
2. ASSESS LIMB/WOUND FOR APPROPRIATE INTERVENTION AND PREPARE PACKING KIT.
3. IF TQ IS REQUIRED (AMPUTATION) THEN PLACE NEW TQ 2-3 INCHES ABOVE WOUND ON VIABLE TISSUE (CONSIDER 2nd TQ ON LEGS).
4. ATTEMPT TO PACKAGE WOUND WITH HAEMOSTATIC GAUZE AND STERILE PLAIN GAUZE THEN ECB TO SECURE WOUND PACKING (CONSIDER PROCEDURAL SEDATION)
5. SLOWLY RELEASE INITIAL TQ A QUARTER TURN, REASSESS FOR BLEEDING FOR 30 SECONDS, REPEATING THIS UNTIL TQ IS LOOSE BUT KEEP IN SITU.
6. IF BLEEDING RESTARTS, THEN RETIGHTEN INITIAL TQ AND REASSESS PATIENT.
7. SEEK FURTHER GUIDANCE FROM REACHBACK REGARDING SECOND ATTEMPT BUT IT SHOULD NOT BE CONSIDERED UNTIL DOUBLE THE TIME HAS PASSED SINCE INITIAL TQ APPLICATION AND IT IS FEASIBLE TO DO SO IN PROLONGED CASUALTY CARE SETTING.

ISOTONIC FLUID RESUSCITATION GUIDANCE

IF PHBT IS NOT AVAILABLE/EXHAUSTED and UNABLE TO DO EMERGENCY DONOR PANEL (EDP) CONSIDER 250ml FLUID BOLUSES FOR SHOCKED PATIENTS AND REASSESS FOR RADIAL PULSE OR ADEQUATE SBP AS SEEN ABOVE.
MAX DOSE OF ISOTONIC FLUIDS: 20ml/kg

POINTS TO NOTE

1. IN THE NON-PERMISSIVE/CARE UNDER FIRE (CUF) ENVIRONMENT, IT IS ACCEPTABLE TO PLACE A TQ HIGH AND TIGHT ON THE LIMB BUT THE PATIENT SHOULD BE MOVED TO A MORE PERMISSIBLE ENVIRONMENT TO ASSESS THE WOUND PROPERLY IF FEASIBLE AND PLACE TQ AS LOW AS VIABLY POSSIBLE.
2. TQ AND WOUND PACKING WILL BE EXTREMELY PAINFUL SO ANALGESIA SHOULD BE CONSIDERED IN ACCORDANCE WITH CONSCIOUS LEVELS AND GUIDELINES.
3. TORSO CAVITIES WILL REQUIRE EXTENSIVE PACKING WHICH IS SELDOM FEASIBLE IN THE PREHOSPITAL SETTING DUE TO RESOURCES SO RECOGNITION AND URGENT CONVEYANCE TO DCS IS PARAMOUNT.
4. A MAIN INDICATION FOR FLUID RESUSCITATION IN PENETRATING THORACIC TRAUMA IS MENTAL STATUS. REDUCED CONSCIOUSNESS SHOULD PROMPT PHBT PROTOCOLS IF FEASIBLE.
5. A PELVIC BINDER SHOULD BE PLACED IF THE MECHANISM INDICATES AND THERE IS CONCERN FOR ANY INTERNAL HAEMORRHAGE (SHOCKED PATIENT, PAIN, PERINEAL BRUISING, URETHRAL BLEEDING, ETC.)
6. A RADIAL PULSE IS SUGGESTIVE OF ADEQUATE PERIPHERAL BLOOD PRESSURE AND CAN BE USED AS A INDICATOR OF FLUID IN THE ABSENCE OF MONITORING EQUIPMENT
7. 1g TXA SHOULD BE ADMINISTERED INTRAVENOUSLY OVER 10 MINUTES AND WITHIN 3 HOURS OF INJURY. IT CAN BE USED FOR PATIENTS WITH SUSPECTED HEAD INJURIES.
8. CALCIUM CHLORIDE CAN CAUSE PERIPHERAL VASCULAR IRRITATION SO SHOULD BE FLUSHED WITH 20ml SALINE.
9. TQ DOWNGRADE IDEALLY SHOULD OCCUR IN UNDER 2 HOURS AND UNABLE TO CONVEY TO DEFINITIVE CARE IN A TIMELY MANNER.
10. ISOTONIC FLUIDS ARE A BETTER CHOICE THAN NOTHING.

Guideline written in accordance with SFM Clinical Standard Operating Guideline 005 'Traumatic Haemorrhage' (2022) and MERT SOP 01 'Blood Product and Fluid Resuscitation in Trauma V3.0' (2022).

References

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